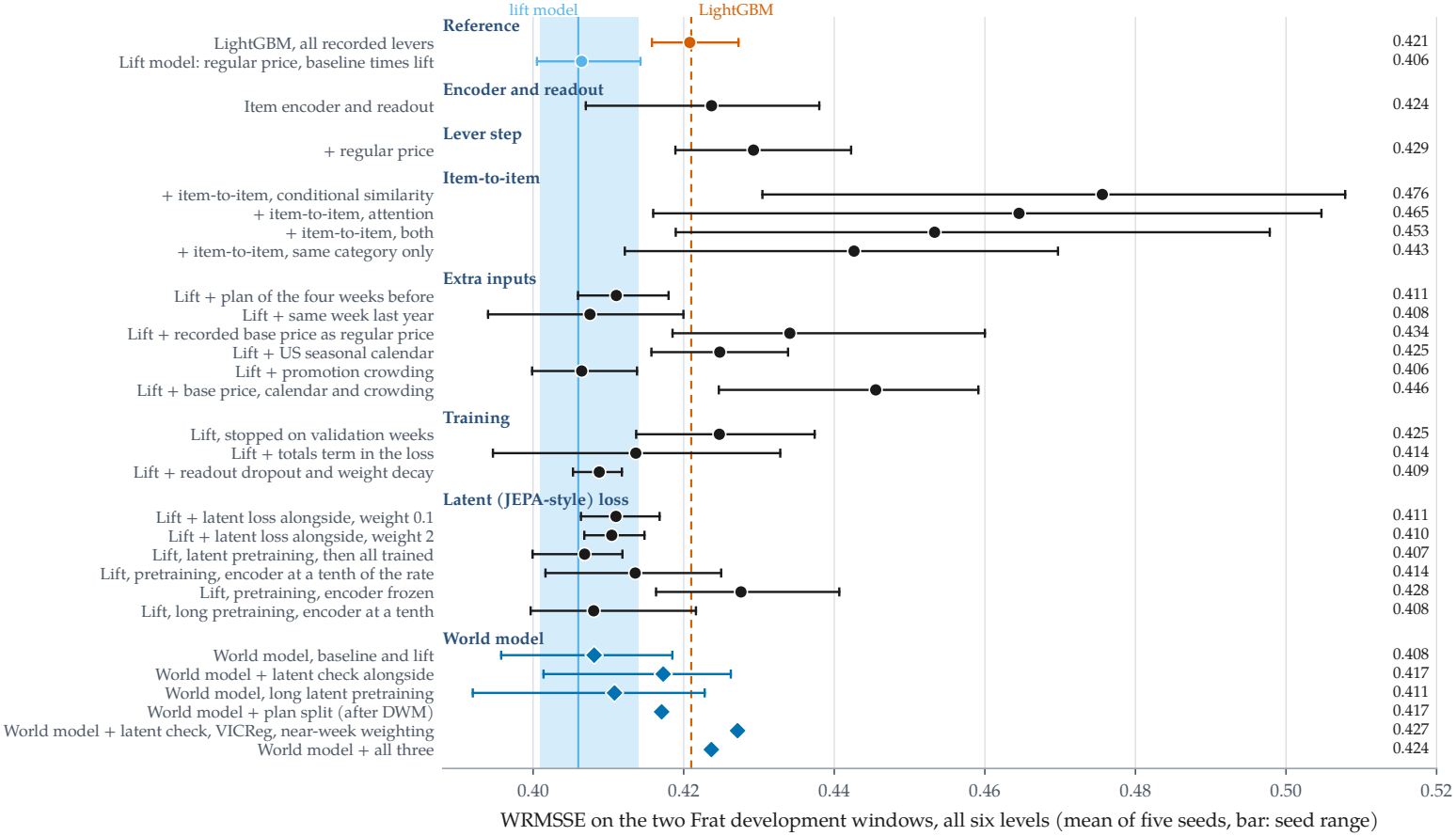
lift model
LightGBM
Reference
LightGBM, all recorded levers
0.421
Lift model: regular price, baseline times lift
0.406
Encoder and readout
Item encoder and readout
0.424
Lever step
+ regular price
0.429
Item-to-item
+ item-to-item, conditional similarity
0.476
+ item-to-item, attention
0.465
+ item-to-item, both
0.453
+ item-to-item, same category only
0.443
Extra inputs
Lift + plan of the four weeks before
0.411
Lift + same week last year
0.408
Lift + recorded base price as regular price
0.434
Lift + US seasonal calendar
0.425
Lift + promotion crowding
0.406
Lift + base price, calendar and crowding
0.446
Training
Lift, stopped on validation weeks
0.425
Lift + totals term in the loss
0.414
Lift + readout dropout and weight decay
0.409
Latent (JEPA-style) loss
Lift + latent loss alongside, weight 0.1
0.411
Lift + latent loss alongside, weight 2
0.410
Lift, latent pretraining, then all trained
0.407
Lift, pretraining, encoder at a tenth of the rate
0.414
Lift, pretraining, encoder frozen
0.428
Lift, long pretraining, encoder at a tenth
0.408
World model
World model, baseline and lift
0.408
World model + latent check alongside
0.417
World model, long latent pretraining
0.411
World model + plan split (after DWM)
0.417
World model + latent check, VICReg, near-week weighting
0.427
World model + all three
0.424
0.40
0.42
0.44
0.46
0.48
0.50
0.52
WRMSSE on the two Frat development windows, all six levels (mean of five seeds, bar: seed range)